

# Docker challenge

## 1. Create a customized Docker image using a Dockerfile.

- Create mkdir app and created index file

```
root@ip-172-31-25-147:~# mkdir myapp
root@ip-172-31-25-147:~# cd myapp
root@ip-172-31-25-147:~/myapp# vi index.html
```

```
<h1>Hello from Customized Docker Image</h1>
~
~
```

- Created vi dockerfile

```
# Use base image
FROM nginx:latest

# Maintainer information
LABEL maintainer="Arun Sai"

# Remove default nginx page
RUN rm -rf /usr/share/nginx/html/*

# Copy custom application files
COPY index.html /usr/share/nginx/html/

# Expose container port
EXPOSE 80

# Start nginx server
CMD ["nginx", "-g", "daemon off;"]
```

- Build Docker Image

```
root@ip-172-31-25-147:~/myapp# docker build -t custom-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/6 : FROM nginx:latest
--> 5aca99593157
Step 2/6 : LABEL maintainer="Arun Sai"
--> Running in 4b180de57949
--> Removed intermediate container 4b180de57949
--> 743c1f59b10a
Step 3/6 : RUN rm -rf /usr/share/nginx/html/*
--> Running in 0beca39f03e3
--> Removed intermediate container 0beca39f03e3
--> cc37a4505763
Step 4/6 : COPY index.html /usr/share/nginx/html/
--> 35dd3021c761
Step 5/6 : EXPOSE 80
--> Running in 2e5a9aec2830
--> Removed intermediate container 2e5a9aec2830
--> fda468c95ac2
Step 6/6 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 71c7af5191bf
--> Removed intermediate container 71c7af5191bf
--> 1112feeaf03e
Successfully built 1112feeaf03e
Successfully tagged custom-nginx:latest
```

➤ Check docker images

```
root@ip-172-31-25-147:~/myapp# docker images
```

| IMAGE                                                                | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|----------------------------------------------------------------------|--------------|------------|--------------|-------|
| 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure:latest    | 371e84608aa7 | 584MB      | 195MB        | U     |
| 928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo:latest | 371e84608aa7 | 584MB      | 195MB        | U     |
| alpine:tomcat:v1                                                     | f61ad2630e2b | 494MB      | 178MB        | U     |
| alpine:latest                                                        | 5b10f432ef3d | 13.1MB     | 3.95MB       |       |
| amazonlinux:latest                                                   | ad882807c6c8 | 227MB      | 59.3MB       |       |
| arunsai17/custom-nginx:v1                                            | 9943b2ca5760 | 238MB      | 63.1MB       |       |
| arunsai17/techie19:v1                                                | 925917d788e0 | 584MB      | 195MB        |       |
| custom-nginx-backup:v1                                               | 22764ef71228 | 584MB      | 195MB        |       |
| custom-nginx:latest                                                  | 1112feeaf03e | 238MB      | 63.1MB       |       |
| custom-nginx:v1                                                      | 371e84608aa7 | 584MB      | 195MB        | U     |
| alpine:tomcat:v1                                                     | 0d70882e6879 | 420MB      | 114MB        |       |

➤ Run Docker Container

```
root@ip-172-31-25-147:~/myapp# docker run -d -p 8080:80 --name mycontainer custom-nginx
7c75bf88854394a1ce26e44c4ec80c85693992544aa17c4cd65c4980029c12b1
root@ip-172-31-25-147:~/myapp#
```

➤ Verify application

← → ↺ ⚠ Not secure 54.226.179.195:8080

# Hello from Customized Docker Image

➤ Check Running Containers stop and remove it

```
root@ip-172-31-25-147:~/myapp# docker ps
```

| CONTAINER ID | IMAGE         | COMMAND                 | CREATED           | STATUS       |
|--------------|---------------|-------------------------|-------------------|--------------|
| 7c75bf888543 | custom-nginx  | "/docker-entrypoint..." | 4 minutes ago     | Up 4 minutes |
| fa883ac62cd6 | tomcat:latest | "catalina.sh run"       | About an hour ago | Up About an  |
| 9210689e1509 | python-app:v1 | "python app.py"         | 3 hours ago       | Up 3 hours   |

```
root@ip-172-31-25-147:~/myapp# docker stop 7c75bf888543
7c75bf888543
root@ip-172-31-25-147:~/myapp# docker rm 7c75bf888543
7c75bf888543
root@ip-172-31-25-147:~/myapp# docker rmi custom-nginx
Untagged: custom-nginx:latest
Deleted: sha256:1112feeaf03ed50d2f2687dddef06cf1f3c2176c735c5f5db54b95e8e6c918b7
Deleted: sha256:fda468c95ac2087ed36d418e7cf60a6c78c5fec01729fab01c44eb8ac1dab024
Deleted: sha256:35dd3021c761ebab211cd28763008c4c8a514dcd4473a619db67c1ed60ca79d9
Deleted: sha256:cc37a450576394ec0eb86eebfeebffcfb9a3aeed600fd36565c6e4b2197e16df
Deleted: sha256:743c1f59b10acf96a2daa67f47e71d63cc968ee69318e04cc00aa67c5f59659e
root@ip-172-31-25-147:~/myapp#
```

## 2. Push the image to Docker Hub.

➤ Created new Docker hub and created new repository

The screenshot shows the Docker Hub web interface. On the left, there's a sidebar with the user profile 'arunsai17' and navigation links: Repositories, Hardened Images, Collaborations, and Settings. The main area displays the repository 'arunsai17/techie19' under the 'General' tab. It shows the repository was created 'less than a minute ago' and has 0 stars and 0 downloads. There are links to 'Add a description' and 'Add a category'.

- Check docker images and change the name

```
root@ip-172-31-25-147:~/docker-nginx# docker images
```

| IMAGE                  | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|------------------------|--------------|------------|--------------|-------|
| amazonlinux:latest     | ad882807c6c8 | 227MB      | 59.3MB       |       |
| custom-nginx:v1        | 925917d788e0 | 584MB      | 195MB        | U     |
| hello-world:latest     | 0e760fdfbc48 | 25.9kB     | 9.49kB       | U     |
| jenkins/jenkins:latest | d566c401e890 | 801MB      | 290MB        | U     |
| nginx:latest           | 5aca99593157 | 241MB      | 66MB         | U     |
| sonarqube:latest       | 35bedac3f40c | 2.52GB     | 1.08GB       |       |
| tomcat:9               | 5f08b8b793ed | 592MB      | 158MB        | U     |
| tomcat:latest          | 247650787b97 | 589MB      | 158MB        |       |
| ubuntu:latest          | f3d28607ddd7 | 160MB      | 45.3MB       |       |

```
root@ip-172-31-25-147:~/docker-nginx# docker tag custom-nginx:v1 arunsail7/techie19:v1
root@ip-172-31-25-147:~/docker-nginx# docker images
```

| IMAGE                  | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|------------------------|--------------|------------|--------------|-------|
| amazonlinux:latest     | ad882807c6c8 | 227MB      | 59.3MB       |       |
| arunsail7/techie19:v1  | 925917d788e0 | 584MB      | 195MB        | U     |
| custom-nginx:v1        | 925917d788e0 | 584MB      | 195MB        | U     |
| hello-world:latest     | 0e760fdfbc48 | 25.9kB     | 9.49kB       | U     |
| jenkins/jenkins:latest | d566c401e890 | 801MB      | 290MB        | U     |
| nginx:latest           | 5aca99593157 | 241MB      | 66MB         | U     |
| sonarqube:latest       | 35bedac3f40c | 2.52GB     | 1.08GB       |       |
| tomcat:9               | 5f08b8b793ed | 592MB      | 158MB        | U     |
| tomcat:latest          | 247650787b97 | 589MB      | 158MB        |       |
| ubuntu:latest          | f3d28607ddd7 | 160MB      | 45.3MB       |       |

- Login with docker hub with URL and password

```
root@ip-172-31-25-147:~/docker-nginx# docker login
```

USING WEB-BASED LOGIN

**Info →** To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: RCMS-RMQN  
 Press ENTER to open your browser or submit your device code here: <https://login.docker.com/activate>

Waiting for authentication in the browser...

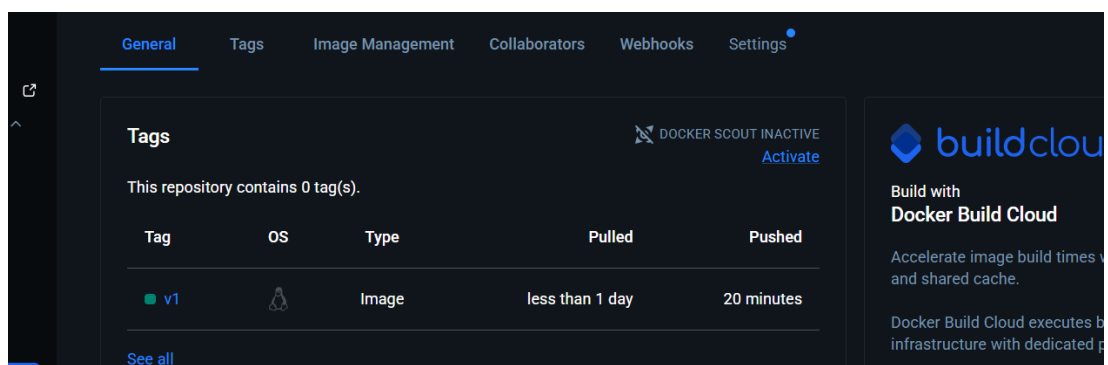
WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.  
 Configure a credential helper to remove this warning. See  
<https://docs.docker.com/go/credential-store/>

Login Succeeded

- Pushed to Docker hub successfully

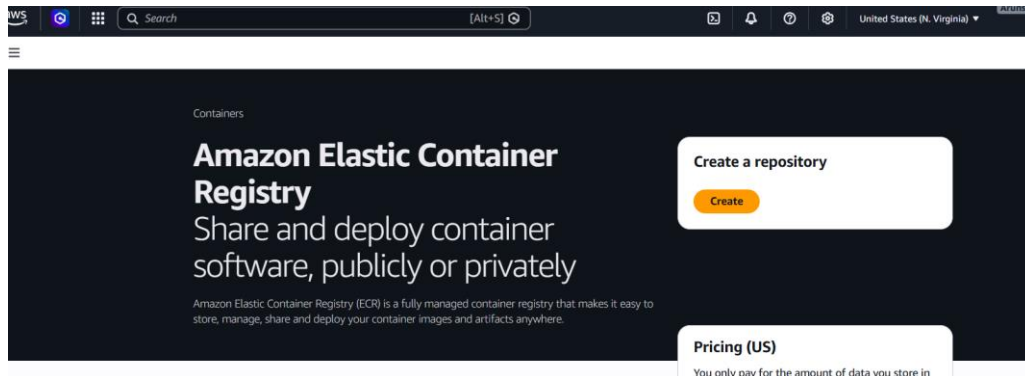
```
root@ip-172-31-25-147:~/docker-nginx# docker push arunsail7/techie19:v1
The push refers to repository [docker.io/arunsail7/techie19]
10d3dbd157fe: Pushed
c7345fb8c651: Mounted from library/amazonlinux
bcf200246380: Pushed
v1: digest: sha256:925917d788e011a8b695385588af86d454c8da06ad703ef598d08392671ebb6d size: 972
```

- Checked in Docker Hub successfully

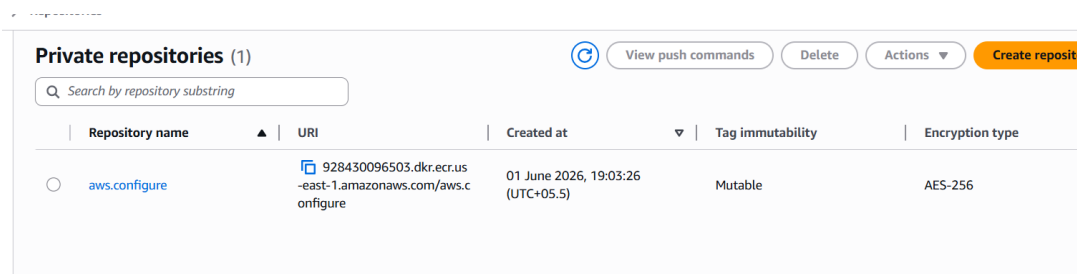


### 3. Push the same image to Amazon ECR.

- Open AWS ECR and create Repo



- Created new repo



- Login with AWS configure cli

```
root@ip-172-31-25-147:~/docker-nginx# cd aws cli
-bash: cd: too many arguments
root@ip-172-31-25-147:~/docker-nginx# cd
root@ip-172-31-25-147:~# aws cli
Command 'aws' not found, but can be installed with:
apt install awscli
root@ip-172-31-25-147:~# apt install awscli
Installing:
awscli
Installing dependencies:
docutils-common libjpeg-turbo8 libpaper2 libwebpmux3 python3-prompt-toolkit
libdeflate0 libjpeg8 libraqm0 python3-awscrt python3-roman-numerals
libgraphite2-3 liblcms2-2 libsharpyuv0 python3-colorama python3-ruamel.yaml
libharfbuzz0b liblerc4 libtiff6 python3-docutils python3-ruamel.yaml.clib
libimagequant0 libopenjp2-7 libwebp7 python3-olefile python3-wcwidth
libjbig0 libpaper-utils libwebpdemux2 python3-pil
Suggested packages:
liblcms2-utils fonts-linuxlibertine texlive-lang-french texlive-latex-recommended
docutils-doc libttf-linux-libertine texlive-latex-base python-pil-doc
```

- After login with secret details and login to docker

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-25-147:~# aws configure
AWS Access Key ID [None]: AKIA5QKV7AB3RFRZF7W2
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
root@ip-172-31-25-147:~# aws ecr get-login-password --region us-east-1 | \
docker login \
--username AWS \
--password-stdin 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure
WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-25-147:~# docker images
```

- Check docker images

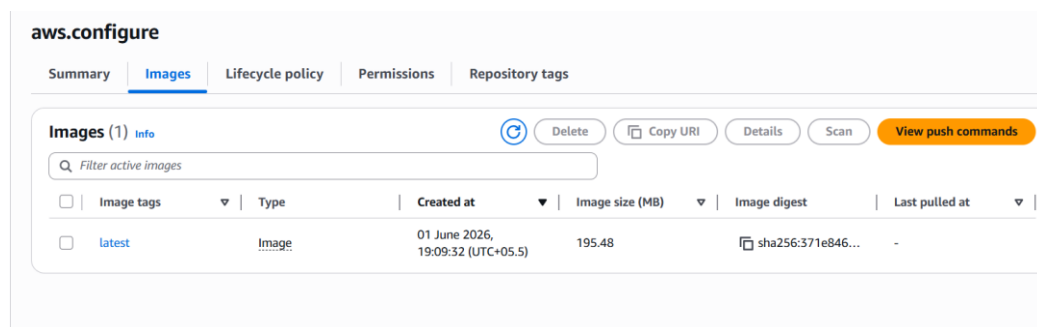
```
login Succeeded
root@ip-172-31-25-147:~# docker images
```

| IMAGE                  | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|------------------------|--------------|------------|--------------|-------|
| amazonlinux:latest     | ad882807c6c8 | 227MB      | 59.3MB       |       |
| arunsail7/techie19:v1  | 925917d788e0 | 584MB      | 195MB        |       |
| custom-nginx:v1        | 371e84608aa7 | 584MB      | 195MB        | U     |
| hello-world:latest     | 0e760fdfbc48 | 25.9kB     | 9.49kB       | U     |
| jenkins/jenkins:latest | d566c401e890 | 801MB      | 290MB        | U     |
| nginx:latest           | 5aca99593157 | 241MB      | 66MB         | U     |
| sonargube:latest       | 35bedac3f40c | 2.52GB     | 1.08GB       |       |
| tomcat:9               | 5f08b8b793ed | 592MB      | 158MB        | U     |
| tomcat:latest          | 247650787b97 | 589MB      | 158MB        |       |
| ubuntu:latest          | f3d28607ddd7 | 160MB      | 45.3MB       |       |

- Now push docker to repo with tag

```
root@ip-172-31-25-147:~# docker tag custom-nginx:v1 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure
root@ip-172-31-25-147:~# docker push 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure
Using default tag: latest
The push refers to repository [928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure]
93a064f8c941: Pushed
c7345fb8c651: Pushed
bcf200246380: Pushed
latest: digest: sha256:371e84608aa78f1a29a9e310e6f485fff9b5700aaba2c4b8f2497dac153829f8 size: 972
root@ip-172-31-25-147:~#
```

- In Repo we can see successfully pushed



#### 4. Provision one EC2 instance using Terraform and install Jenkins.

- Install Terraform

```
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform -version
Terraform v1.15.5
on linux_amd64
+ provider registry.terraform.io/hashicorp/aws v6.48.0
ubuntu@ip-172-31-25-147:~/terraform-jenkins$
```

- Configure AWS Credentials

```
ubuntu@ip-172-31-25-147:~$ aws configure
AWS Access Key ID [*****F7W2]: AKIA5QKV7AB3RFRZF7W2
AWS Secret Access Key [*****eh8k]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json
ubuntu@ip-172-31-25-147:~$ mkdir terraform-jenkins
ubuntu@ip-172-31-25-147:~$ cd terraform-jenkins
```

➤ Create Terraform Files

```
provider registry.terraform.io/hashicorp/aws v0.40.0
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ vi main.tf
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ vi userdata.sh
ubuntu@ip-172-31-25-147:~/terraform-jenkins$
```

➤ Main data script

```
provider "aws" {
  region = "us-east-1"
}

data "aws_ami" "ubuntu" {
  most_recent = true

  owners = ["099720109477"]

  filter {
    name = "name"
    values = ["ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-*"]
  }
}

resource "aws_security_group" "jenkins_sg" {
  name = "jenkins-security-group"

  ingress {
    from_port = 22
    to_port   = 22
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port = 8080
    to_port   = 8080
  }
}
```

➤ User Data Script

```
#!/bin/bash

apt update -y

apt install openjdk-21-jdk -y

wget -O /usr/share/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key

echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" \
| tee /etc/apt/sources.list.d/jenkins.list > /dev/null

apt update -y

apt install jenkins -y

systemctl enable jenkins
systemctl start jenkins
```

➤ Terraform apply

```
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform state list
aws_security_group.jenkins_sg
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform apply
aws_security_group.jenkins_sg: Refreshing state... [id=sg-0bde2212bcacc50bb]

Terraform used the selected providers to generate the following execution plan. Resource actions are
+ create

Terraform will perform the following actions:

# aws_instance.jenkins_server will be created
+ resource "aws_instance" "jenkins_server" {
  + ami           = "ami-0c02fb55956c7d316"
  + arn           = (known after apply)
}
```

➤ Completed Success Terraform

```
Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.jenkins_server: Creating...
aws_instance.jenkins_server: Still creating... [00m10s elapsed]
aws_instance.jenkins_server: Creation complete after 16s [id=i-05a23942cd494e6d0]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform state show aws_instance.jenkins_server
# aws_instance.jenkins_server:
resource "aws_instance" "jenkins_server" {
  ami              = "ami-0c02fb55956c7d316"
  arn              = "arn:aws:ec2:us-east-1:928430096503:instance/i-05a23942cd494e6d0"
  associate_public_ip_address = true
}
```

➤ The output public id new EC2 server

```
public_ip = "3.88.104.248"
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ ls
main.tf  terraform.tfstate  terraform.tfstate.backup  userdata.sh
```

➤ New Server created of EC2

| <input type="checkbox"/> | Name           | Instance ID                         | Instance state | Instance type  | Status check      | Availability Zone |
|--------------------------|----------------|-------------------------------------|----------------|----------------|-------------------|-------------------|
| <input type="checkbox"/> | Docker-server  | <a href="#">i-093ae5d78efbc2b15</a> | Running        | c7i-flex.large | 3/3 checks passed | us-east-1c        |
| <input type="checkbox"/> | Jenkins-Server | <a href="#">i-04a7f8dcfed325643</a> | Running        | t3.micro       | 3/3 checks passed | us-east-1c        |

➤ Now check new EC2 server java version

```
ubuntu@ip-172-31-22-87:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-24.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-24.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-22-87:~$
```

➤ Manually installed Jenkins

```
ubuntu@ip-172-31-22-87:~$ wget https://get.jenkins.io/debian-stable/jenkins_2.555.2_all.deb
--2026-06-04 10:05:33-- https://get.jenkins.io/debian-stable/jenkins_2.555.2_all.deb
Resolving get.jenkins.io (get.jenkins.io)... 128.24.70.119, 2603:1030:408:3::2bd
Connecting to get.jenkins.io (get.jenkins.io)|128.24.70.119|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://ftp-nyc.osuosl.org/pub/jenkins/debian-stable/jenkins_2.555.2_all.deb [following]
--2026-06-04 10:05:33-- https://ftp-nyc.osuosl.org/pub/jenkins/debian-stable/jenkins_2.555.2_all.deb
Resolving ftp-nyc.osuosl.org (ftp-nyc.osuosl.org)... 64.50.233.100, 2600:3404:200:237::2
Connecting to ftp-nyc.osuosl.org (ftp-nyc.osuosl.org)|64.50.233.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 99442780 (95M) [application/vnd.debian.binary-package]
Saving to: 'jenkins_2.555.2_all.deb'

jenkins_2.555.2_all.deb      100%[=====]
2026-06-04 10:05:34 (161 MB/s) - 'jenkins_2.555.2_all.deb' saved [99442780/99442780]

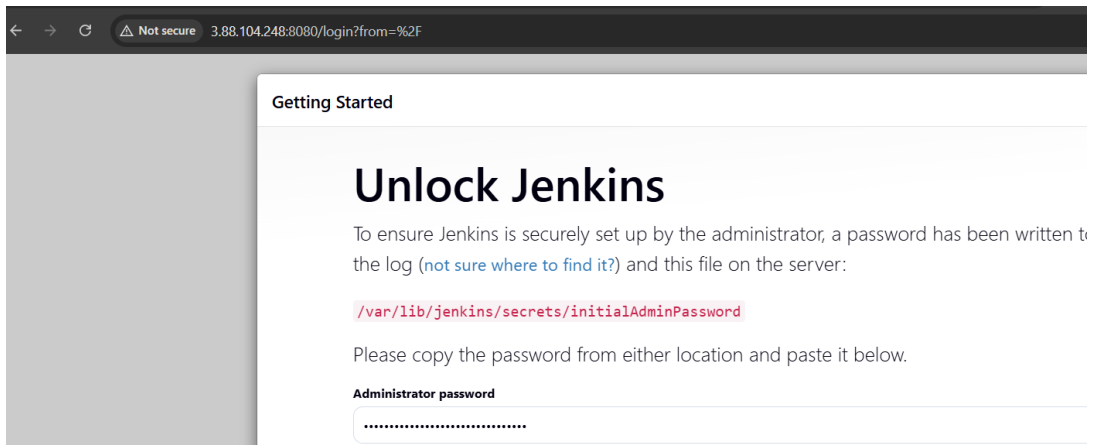
ubuntu@ip-172-31-22-87:~$ ls -lh jenkins_2.555.2_all.deb
-rw-rw-r-- 1 ubuntu ubuntu 95M May 13 16:19 jenkins_2.555.2_all.deb
```

➤ Now start Jenkins and check status Jenkins

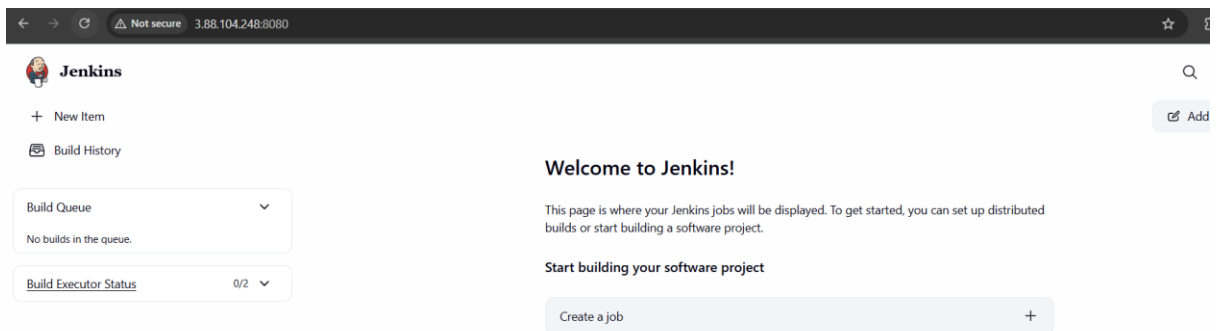
```
needrestart is being skipped since dpkg has failed
ubuntu@ip-172-31-22-87:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-22-87:~$ sudo systemctl status jenkins
• jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-06-04 10:06:37 UTC; 1min 16s ago
     Main PID: 5840 (java)
       Tasks: 37 (limit: 1013)
      Memory: 310.6M (peak: 344.6M)
         CPU: 20.417s
    CGroup: /system.slice/jenkins.service
            └─5840 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/v

Jun 04 10:06:32 ip-172-31-22-87 jenkins[5840]: [LF]> This may also be found at: /var/lib/jenkins/secrets
Jun 04 10:06:32 ip-172-31-22-87 jenkins[5840]: [LF]>
Jun 04 10:06:32 ip-172-31-22-87 jenkins[5840]: [LF]> *****
Jun 04 10:06:32 ip-172-31-22-87 jenkins[5840]: [LF]> *****
Jun 04 10:06:32 ip-172-31-22-87 jenkins[5840]: [LF]> *****
Jun 04 10:06:37 ip-172-31-22-87 jenkins[5840]: 2026-06-04 10:06:37.206+0000 [id=32]      INFO      j
Jun 04 10:06:37 ip-172-31-22-87 jenkins[5840]: 2026-06-04 10:06:37.258+0000 [id=24]      INFO      h
Jun 04 10:06:37 ip-172-31-22-87 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Ser
Jun 04 10:06:37 ip-172-31-22-87 jenkins[5840]: 2026-06-04 10:06:37.498+0000 [id=50]      INFO      h
Jun 04 10:06:37 ip-172-31-22-87 jenkins[5840]: 2026-06-04 10:06:37.499+0000 [id=50]      INFO      h
lines 1-20/20 (END)
```

➤ Now with Public Id of new EC2 with port no 8080



➤ Jenkins install plugins and done



## 5. Create one Jenkins job to build and push the Docker image to Docker Hub.

- Pull Jenkins and installed it

```
root@ip-172-31-25-147:~# docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
3bdbd616c406: Pull complete
cebc1e77ceba: Pull complete
307f8152a55e: Pull complete
5834ae4b8774: Pull complete
6b35bb76b8c8: Pull complete
801814f62c99: Pull complete
c2314d42ccal: Pull complete
76a355a69c2e: Pull complete
9014a3401c36: Pull complete
4c140080734e: Pull complete
dd00b4ea1f64: Pull complete
ae218d1cfa6e: Pull complete
653a4e57bccf: Download complete
Digest: sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
root@ip-172-31-25-147:~# docker run -d \
--name jenkins \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
jenkins/jenkins:lts
0fcaa0203d459266682dddee245ffa53d654c5fd1df005a4d19c78dbd32f7b6c
```

- Checked running containers Jenkins

```
root@ip-172-31-25-147:~# docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS
0fcaa0203d45   jenkins/jenkins:lts  "/usr/bin/tini -- /u..."  14 seconds ago  Up 13 seconds  0.0.0.0:8080->8080,0.0.0.0:50000->50000/tcp
root@ip-172-31-25-147:~# docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
001eb9a1cea9475f9a62cb93693b76fa
```

- Update it and install docker again

```

root@0fcaa0203d45:/# apt update
apt install docker.io -y
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [195 kB]
Fetched 10.1 MB in 1s (12.5 MB/s)
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
Upgrading:
  libcap2  libsystemd0  libudev1

Installing:
  docker.io

```

- Checked docker version

```

exit
root@ip-172-31-25-147:~# docker exec -it jenkins docker version
Client:
 Version:           26.1.5+dfsg1
 API version:       1.45
 Go version:        go1.24.4
 Git commit:        a72d7cd
 Built:             Sat May  9 11:34:09 2026
 OS/Arch:           linux/amd64
 Context:           default

```

- Connect Jenkins with root and update it restart Jenkins

```

root@ip-172-31-25-147:~# docker exec -u root -it jenkins bash
root@0fcaa0203d45:/# apt update
apt install docker.io -y
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
21 packages can be upgraded. Run 'apt list --upgradable' to see them.
docker.io is already the newest version (26.1.5+dfsg1-9+b13).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 21
root@0fcaa0203d45:/# exit
exit
root@ip-172-31-25-147:~# docker restart jenkins
jenkins
root@ip-172-31-25-147:~# docker exec -it jenkins docker version
Client:
 Version:           26.1.5+dfsg1
 API version:       1.45

```

- Give permissions to docker and checked version

```

al/run/docker.sock: connect: permission denied
root@ip-172-31-25-147:~# chmod 666 /var/run/docker.sock
root@ip-172-31-25-147:~# docker exec -it jenkins docker version
Client:
 Version:           26.1.5+dfsg1
 API version:       1.45
 Go version:        go1.24.4
 Git commit:        a72d7cd
 Built:             Sat May  9 11:34:09 2026
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          29.1.3

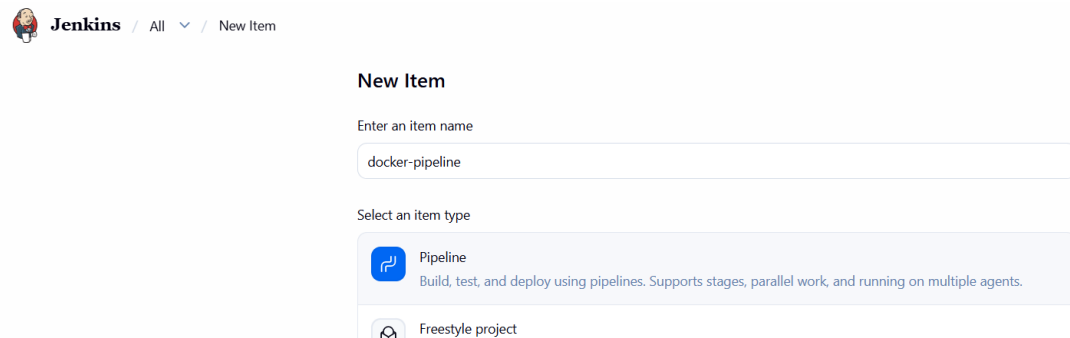
```

- Created index and docker file and save it

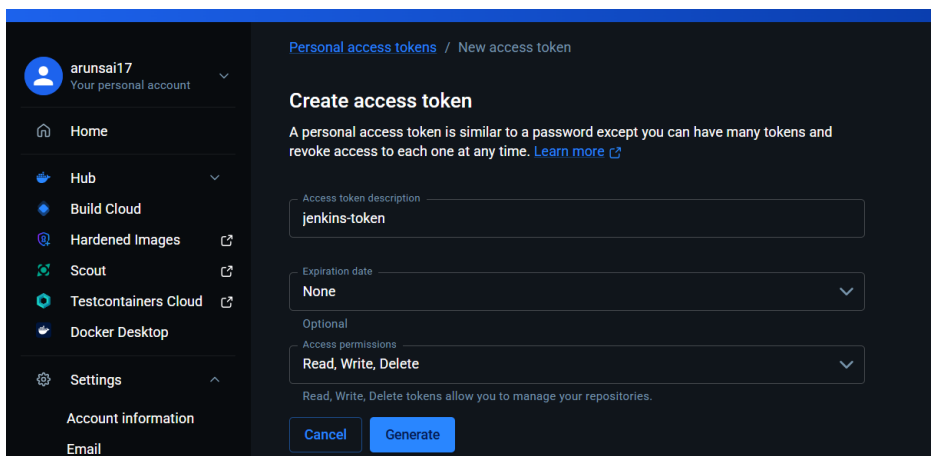
```
cd docker-project
root@ip-172-31-25-147:~/docker-project# vi index.html
root@ip-172-31-25-147:~/docker-project# vi Dockerfile
root@ip-172-31-25-147:~/docker-project# ls
Dockerfile  index.html
root@ip-172-31-25-147:~/docker-project# docker build -t test-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/2 : FROM nginx:latest
--> 5aca99593157
Step 2/2 : COPY index.html /usr/share/nginx/html/index.html
--> 1ef27cf52923
Successfully built 1ef27cf52923
Successfully tagged test-nginx:latest
root@ip-172-31-25-147:~/docker-project#
```

- Open Jenkins URL and done with configuration and created a pipeline



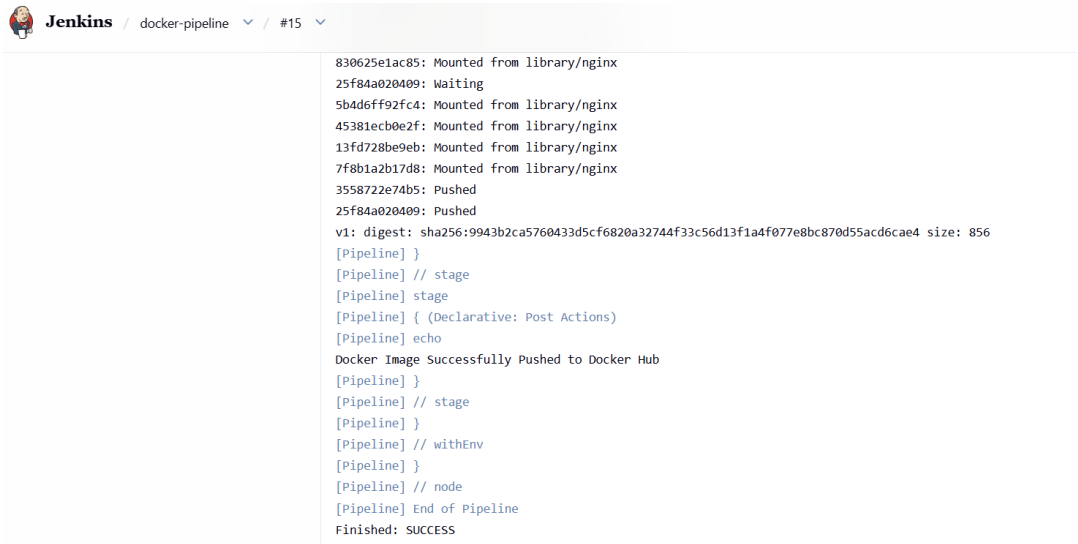
- In Docker Hub Generate Token



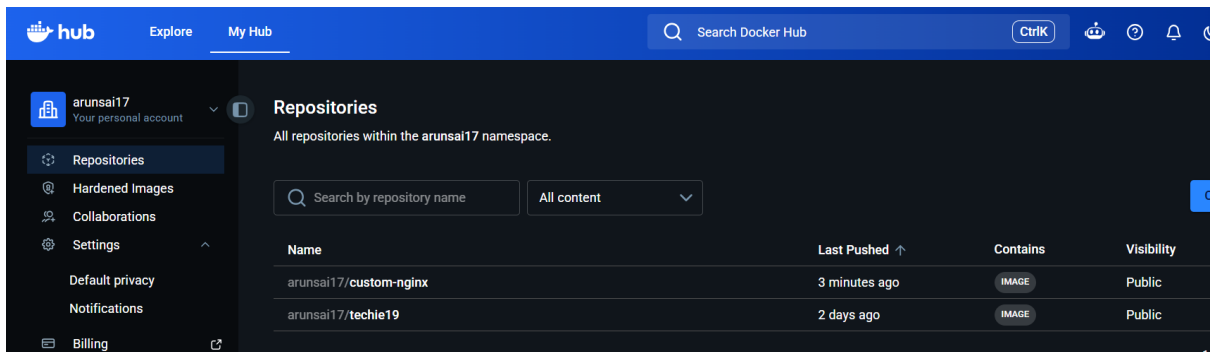
- In Jenkins managing open credentials and create new creds for docker



## ➤ Success Build



## ➤ Check in Docker Hub Repo



- Source: <https://github.com/betawins/Python-app.git>
- Source Codes: <https://github.com/betawins/docker-tasks.git>
- From the frontend source code, write a Dockerfile, build a Docker image, run it, and push that image to your Docker registry.
- Docker version and git version

```

root@ip-172-31-25-147:~# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
root@ip-172-31-25-147:~# git --version
git version 2.53.0
root@ip-172-31-25-147:~# git clone https://github.com/betawins/docker-tasks.git
Cloning into 'docker-tasks'...
remote: Enumerating objects: 146, done.
remote: Counting objects: 100% (146/146), done.
remote: Compressing objects: 100% (123/123), done.
remote: Total 146 (delta 12), reused 141 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (146/146), 22.11 MiB | 80.56 MiB/s, done.
Resolving deltas: 100% (12/12), done.

```

### ➤ Create Dockerfile

```

root@ip-172-31-25-147:~/docker-tasks# cd ~/docker-tasks/Frontend_based_source
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# ls
index.html  javascript.js  style.css  todayDeal.js
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# vi Dockerfile
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# 

```

```

FROM nginx:latest

COPY . /usr/share/nginx/html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

```

### ➤ Build Frontend Image

```

root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker build -t frontend-app
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 44.03kB
Step 1/4 : FROM nginx:latest
--> 5aca99593157
Step 2/4 : COPY . /usr/share/nginx/html
--> d329a31108f1
Step 3/4 : EXPOSE 80
--> Running in dbbf9f971ae3
--> Removed intermediate container dbbf9f971ae3
--> 70db818e61c3
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
--> Running in a005de9cdb16
--> Removed intermediate container a005de9cdb16
--> 6647e688c3af
Successfully built 6647e688c3af
Successfully tagged frontend-app:latest
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# 

```

### ➤ Run Frontend Container

```

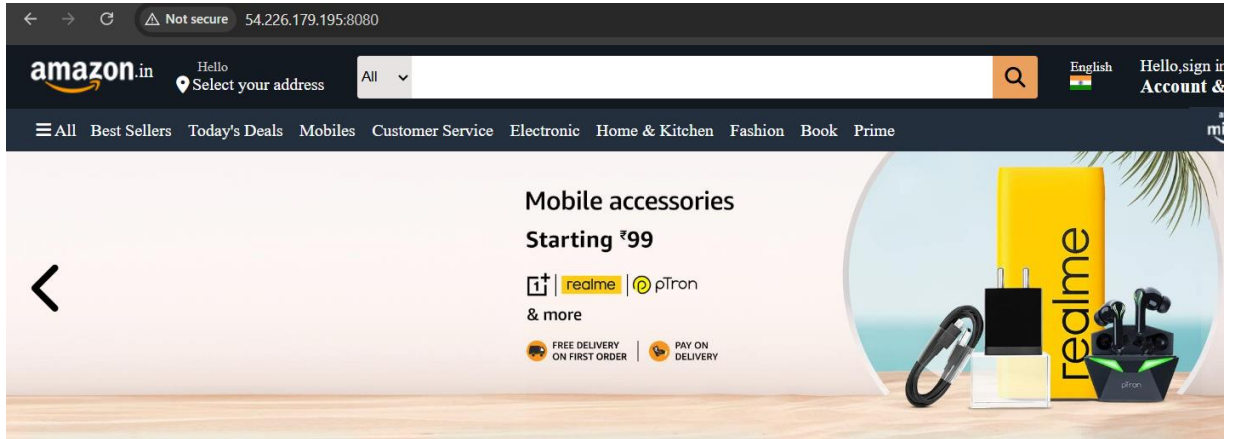
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker run -d -p 8080:80
26c8386b79c2ad401bc2bd306de419c82099de6b44df7729619c164857b4e898
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# 

```

➤ Verify Container

```
26c8386b79c2ad401bc2bd306de419c82099de6b44d17729619c164837b4e698
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
26c8386b79c2   frontend-app   "/docker-entrypoint..." About a minute ago Up About a minute
fa883ac62cd6   tomcat:latest  "catalina.sh run"        2 hours ago   Up 2 hours
```

➤ Access in Browser



➤ Verify Running Container

```
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
26c8386b79c2   frontend-app   "/docker-entrypoint..." 6 minutes ago   Up 6 minutes   0.0.0.0:8080->80/tcp, [:::
fa883ac62cd6   tomcat:latest  "catalina.sh run"        2 hours ago   Up 2 hours     0.0.0.0:8086->8080/tcp, [:::
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source#
```

➤ Login in

```
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker login
Authenticating with existing credentials... [Username: arunsail17]

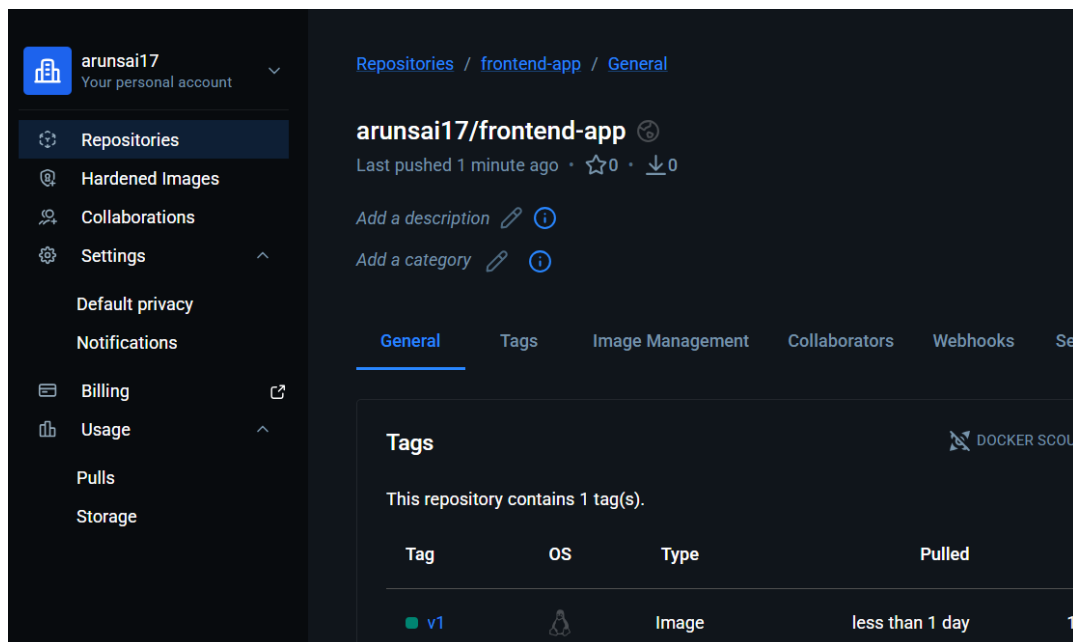
Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

➤ Now Push to Docker Hub

```
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker tag frontend-app arunsail17/frontend-app:v1
root@ip-172-31-25-147:~/docker-tasks/Frontend_based_source# docker push arunsail17/frontend-app:v1
The push refers to repository [docker.io/arunsail17/frontend-app]
5431d0092ffd: Mounted from arunsail17/custom-nginx
b4a248c845e5: Mounted from arunsail17/custom-nginx
7f8b1a2b17d8: Mounted from arunsail17/custom-nginx
45381ecb0e2f: Mounted from arunsail17/custom-nginx
13fd728be9eb: Mounted from arunsail17/custom-nginx
```

➤ Verify on Docker Hub



- From the Java-based source code, write a Dockerfile, build, run, and push the image to the Docker registry.

➤ Java Source Folder

```
root@ip-172-31-25-147:~# cd ~/docker-tasks/Java_based_source
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# ls
pom.xml  src
```

➤ Install Java and Maven

```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# apt install openjdk-17-jdk maven -y
openjdk-17-jdk is already the newest version (17.0.19+10-1~26.04.2).
Installing:
  maven

Installing dependencies:
  libaopalliance-java  libcommons-lang3-java  libhttpclient-java  libmaven3-
  libapache-pom-java  libcommons-logging-java  libhttpcore-java  libplexus-
  libasm-java  libcommons-parent-java  libjansi-java  libplexus-
  libatinject-jsr330-api-java  liberror-prone-java  libjsoup-java  libplexus-
  libcdi-api-java  libgeronimo-annotation-1.3-spec-java  libjsr305-java  libplexus-
  libcommons-cli-java  libgeronimo-interceptor-3.0-spec-java  libmaven-parent-java  libplexus-
  libcommons-codec-java  libguava-java  libmaven-resolver-java  libplexus-
  libcommons-io-java  libguice-java  libmaven-shared-utils-java  libsisu-in
```

➤ Verify Installation

```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1~26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1~26.04.2-Ubuntu, mixed mode, sharing)
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# mvn -version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 17.0.19, vendor: Ubuntu, runtime: /usr/lib/jvm/java-17-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-1006-aws", arch: "amd64", family: "unix"
root@ip-172-31-25-147:~/docker-tasks/Java_based_source#
```

➤ Build Java Application

```
OS name: "linux", version: "7.0.0-1006-aws", arch: "amd64", family: "unix"
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# mvn clean package
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-starter-
.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-starter-
pom (8.1 kB at 16 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-dependen
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-dependen
(127 kB at 1.9 MB/s)
```

### ➤ Verify JAR File

```
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/guava/guava/19.0/guava-19.0.jar (2.3 MB at 8.
[INFO] Replacing main artifact with repackaged archive
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 25.914 s
[INFO] Finished at: 2026-06-03T13:12:59Z
[INFO] -----
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# ls target
classes generated-sources maven-archiver maven-status spring-boot-web.jar spring-boot-web.jar.original
root@ip-172-31-25-147:~/docker-tasks/Java_based_source#
```

### ➤ Create Dockerfile

```
classes generated-sources maven-archiver maven-status spring-boot-we
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# vi Dockerfile
root@ip-172-31-25-147:~/docker-tasks/Java_based_source#
```

```
FROM eclipse-temurin:17

WORKDIR /app

COPY target/spring-boot-web.jar app.jar

EXPOSE 8080

CMD ["java", "-jar", "app.jar"]
```

### ➤ Build Docker Image

```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# vi Dockerfile
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker build -t java-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 24.95MB
Step 1/5 : FROM eclipse-temurin:17
17: Pulling from library/eclipse-temurin
0179dfb07a7d: Pulling fs layer
a32c42059121: Pulling fs layer
1074d15b60c9: Pulling fs layer
f2355d8d10e0: Pulling fs layer
f2355d8d10e0: Download complete
```

➤ Successful Build it

```
Step 2/5 : WORKDIR /app
---> Running in 090f2173bbec
---> Removed intermediate container 090f2173bbec
---> 5ffdd8318256
Step 3/5 : COPY target/spring-boot-web.jar app.jar
---> d394b1684775
Step 4/5 : EXPOSE 8080
---> Running in 91408ef929aa
---> Removed intermediate container 91408ef929aa
---> 665efc5a78cf
Step 5/5 : CMD ["java", "-jar", "app.jar"]
---> Running in 48411bf26654
---> Removed intermediate container 48411bf26654
---> a908ed85ee6d
Successfully built a908ed85ee6d
Successfully tagged java-app:latest
root@ip-172-31-25-147:~/docker-tasks/Java_based_source#
```

➤ Run Java Container and Verify Running Container

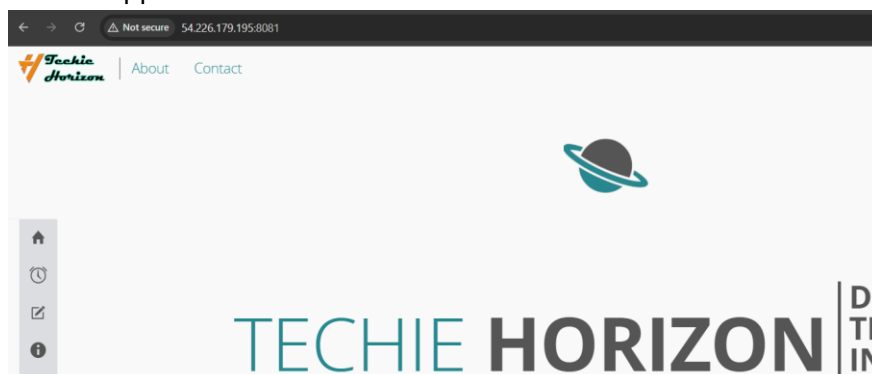
```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker run -d -p 8081:8080 --name java-container
01969ac6b532622a1990c0b833269fec3f5e4136b7b8f858290cd4aff690d16a
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
01969ac6b532   java-app      "/_cacert entrypoint..." 10 seconds ago Up 9 seconds  0.0.0.0:8081-
26c8386b79c2   frontend-app  "/docker-entrypoint..." 22 minutes ago Up 22 minutes  0.0.0.0:8080-
fa883ac62cd6   tomcat:latest "catalina.sh run"         2 hours ago   Up 2 hours    0.0.0.0:8086-
```

➤ Check Logs If Needed

```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker logs java-container
:: Spring Boot :: (v2.2.4.RELEASE)

2026-06-03 13:21:00.216 INFO 1 --- [main] com.hussain.StartApplication
started by root in /app)
2026-06-03 13:21:00.220 INFO 1 --- [main] com.hussain.StartApplication
2026-06-03 13:21:01.669 INFO 1 --- [main] o.s.b.w.embedded.tomcat.TomcatWeb
2026-06-03 13:21:01.683 INFO 1 --- [main] o.apache.catalina.core.StandardSe
2026-06-03 13:21:01.683 INFO 1 --- [main] org.apache.catalina.core.Standard
2026-06-03 13:21:01.760 INFO 1 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/
2026-06-03 13:21:01.761 INFO 1 --- [main] o.s.web.context.ContextLoader
2026-06-03 13:21:02.049 INFO 1 --- [main] o.s.s.concurrent.ThreadPoolTaskEx
2026-06-03 13:21:02.131 INFO 1 --- [main] o.s.b.a.w.s.WelcomePageHandlerMap
2026-06-03 13:21:02.296 INFO 1 --- [main] o.s.b.w.embedded.tomcat.TomcatWeb
2026-06-03 13:21:02.301 INFO 1 --- [main] com.hussain.StartApplication
```

➤ Access Application



➤ Login and verify docker images

```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker login
Authenticating with existing credentials... [Username: arunsail17]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker images
```

| IMAGE                                                                | ID           | DI |
|----------------------------------------------------------------------|--------------|----|
| 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure:latest    | 371e84608aa7 |    |
| 928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo:latest | 371e84608aa7 |    |
| alpine-tomcat:v1                                                     | f61ad2630e2b |    |
| alpine:latest                                                        | 5b10f432ef3d |    |
| amazonlinux:latest                                                   | ad882807c6c8 |    |
| arunsail7/custom-nginx:v1                                            | 9943b2ca5760 |    |
| arunsail7/techie19:v1                                                | 925917d788e0 |    |
| custom-nginx-backup:v1                                               | 22764ef71228 |    |
| custom-nginx:v1                                                      | 371e84608aa7 |    |
| eclipse-temurin:17                                                   | b04a8c5d46e2 |    |
| eclipse-temurin:17-jre                                               | 0d79988c6879 |    |
| frontend-app:latest                                                  | 6647e688c3af |    |
| hello-world:latest                                                   | 0e760fdfbc48 |    |
| java-app:latest                                                      | a908ed85ee6d |    |

➤ Now Push Image to Docker Hub

```
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker tag java-app arunsail7/java-app:v1
root@ip-172-31-25-147:~/docker-tasks/Java_based_source# docker push arunsail7/java-app:v1
The push refers to repository [docker.io/arunsail7/java-app]
1d63b8c43acc: Pushed
6f5c5aa4e145: Mounted from library/ubuntu
1c24335ddd46: Mounted from library/ubuntu
1074d15b60c9: Mounted from library/eclipse-temurin
a32c42059121: Mounted from library/eclipse-temurin
9822af80a281: Pushed
0179dfb07a7d: Mounted from library/eclipse-temurin
f2355d8d10e0: Mounted from library/eclipse-temurin
v1: digest: sha256:a908ed85ee6d2f9d9485866e592eb9c5da8495e90b3d15d193a017be978751ce size: 2053
root@ip-172-31-25-147:~/docker-tasks/Java_based_source#
```

➤ Verify on Docker Hub

The screenshot shows the Docker Hub interface for the repository 'arunsail7/java-app'. The left sidebar contains navigation links: Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, and Storage. The main content area shows the repository details, including the name 'arunsail7/java-app', the last pushed time '1 minute ago', and options to 'Add a description' and 'Add a category'. Below this, there are tabs for 'General', 'Tags', 'Image Management', 'Collaborators', 'Webhooks', and 'Settings'. The 'Tags' tab is selected, showing a table of tags. The table has columns for 'Tag', 'OS', 'Type', 'Pulled', and 'Pushed'. A single tag 'v1' is listed with OS 'linux', Type 'Image', Pulled 'less than 1 day', and Pushed '1 minute'. A 'DOCKER SCOUT INACTIVE' banner is also visible.

| Tag | OS    | Type  | Pulled          | Pushed   |
|-----|-------|-------|-----------------|----------|
| v1  | linux | Image | less than 1 day | 1 minute |

- From the Node.js-based source code, write a Dockerfile, build with tag v1, run, and push it to the Docker registry.

➤ Go to Node.js Source Folder and create Docker File

```
root@ip-172-31-25-147:~/docker-tasks# cd ~/docker-tasks/NodeJs_based_source
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# ls
package-lock.json package.json public src
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# vi Dockerfile
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source#
```

```
FROM node:18

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY . .

EXPOSE 3000

CMD ["npm", "start"]
```

➤ Build Docker Image with v1 Tag

```
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# vi Dockerfile
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# docker build -t node-app:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   4.17MB
Step 1/7 : FROM node:18
18: Pulling from library/node
37927ed901b1: Pulling fs layer
Step 6/7 : EXPOSE 3000
--> Running in 009e99803946
--> Removed intermediate container 009e99803946
--> f406dd0fad7c
Step 7/7 : CMD ["npm", "start"]
--> Running in 3d3b11198434
--> Removed intermediate container 3d3b11198434
--> 30cfe7bf3d23
Successfully built 30cfe7bf3d23
Successfully tagged node-app:v1
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source#
```

➤ Run Node Container

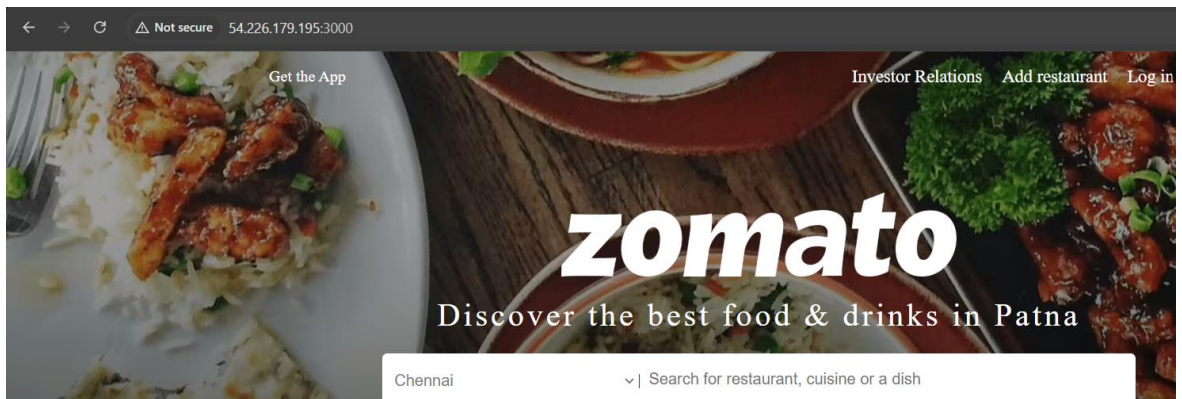
```
---> 30cfe7bf3d23
Successfully built 30cfe7bf3d23
Successfully tagged node-app:v1
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# docker run -d -p 3000:3000 --name node-cont
ce5b164011559321889e91b8fc25f8ec7cf9324fe66f4ce3951f2d926047a2d5
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source#
```

i-093ae5d78efbc2b15 (Docker-server)

➤ Verify Running Container

```
ce5b164011559321889e91b8fc25f8ec7cf9324fe66f4ce3951f2d926047a2d5
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
ce5b16401155   node-app:v1    "docker-entrypoint.s..." About a minute ago Up About a
01969ac6b532   java-app      "/_cacert_entrypoin..." 30 minutes ago Up 30 minut
26c8386b79c2   frontend-app  "/docker-entrypoint...." 52 minutes ago Up 52 minut
fa883ac62cd6   tomcat:latest "catalina.sh run"        3 hours ago   Up 3 hours
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source#
```

➤ Access Node Application



### ➤ Check logs:

```
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# docker logs node-container

> zomato_clone@0.1.0 start
> react-scripts start

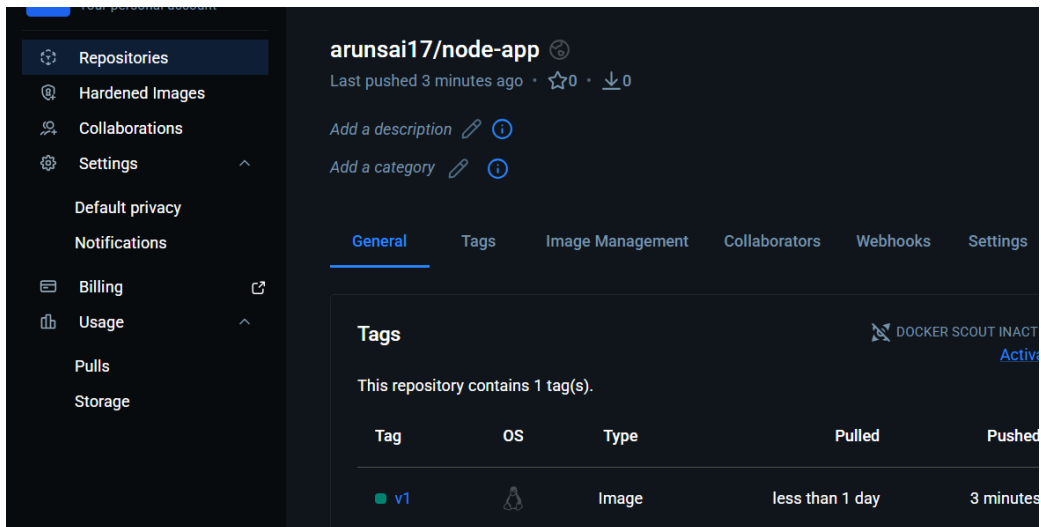
Browserslist: caniuse-lite is outdated. Please run:
  npx update-browserslist-db@latest
  Why you should do it regularly: https://github.com/browserslist/update-db#readme
(node:26) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSe
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:26) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBefore
n.
Starting the development server...

Compiled successfully!
```

### ➤ Push Image to Docker Hub

```
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# docker tag node-app:v1 arunsail17/node-
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source# docker push arunsail17/node-app:v1
The push refers to repository [docker.io/arunsail17/node-app]
4b7de5268354: Pushed
37927ed901b1: Mounted from library/node
79b2f47ad444: Mounted from library/node
3697be50c98b: Mounted from library/node
461077a72fb7: Mounted from library/node
3e6b9d1a9511: Mounted from library/node
e23f099911d6: Mounted from library/node
cda7f44f2bdd: Mounted from library/node
c6b30c3f1696: Mounted from library/node
7fef54d0650d: Pushed
a14e1f51aa8b: Pushed
0cb5c4d20132: Pushed
v1: digest: sha256:30cfe7bf3d23b734f2a0c1dd720b1c77fcea695823699077f29185a2cdfe942f size: 2936
root@ip-172-31-25-147:~/docker-tasks/NodeJs_based_source#
```

### ➤ Verify in Docker Hub



- Write a docker-compose file to set up WordPress with a MySQL database.

- Create Project Folder and created vi docker-compose file

```
root@ip-172-31-25-147:~# mkdir wordpress-compose
root@ip-172-31-25-147:~# cd wordpress-compose
root@ip-172-31-25-147:~/wordpress-compose# vi docker-compose.yml
root@ip-172-31-25-147:~/wordpress-compose#
```

- Create docker-compose.yml File

```
version: '3'
services:
  db:
    image: mysql:5.7
    volumes:
      - db_data:/var/lib/mysql
    restart: always
    environment:
      - MYSQL_ROOT_PASSWORD=somewordpress
      - MYSQL_DATABASE=wordpress
      - MYSQL_USER=wordpress
      - MYSQL_PASSWORD=wordpress
  wordpress:
    depends_on:
      - db
    image: wordpress:latest
    ports:
      - "8000:80"
    restart: always
    environment:
      - WORDPRESS_DB_HOST=db:3306
      - WORDPRESS_DB_USER=wordpress
      - WORDPRESS_DB_PASSWORD=wordpress
      - WORDPRESS_DB_NAME=wordpress
volumes:
  db_data:
```

- Start Containers

```

root@ip-172-31-25-147:~/wordpress-compose# vi docker-compose.yml
root@ip-172-31-25-147:~/wordpress-compose# docker compose up -d
WARN[0000] /root/wordpress-compose/docker-compose.yml: the attribute `version` is obsolete, it will be ignored
[+] Running 40/40
  ✓ Wordpress Pulled
  ✓ db Pulled

[+] Running 4/4
  ✓ Network wordpress-compose_default      Created
  ✓ Volume wordpress-compose_db_data      Created
  ✓ Container wordpress-compose-db-1      Started
  ✓ Container wordpress-compose-wordpress-1 Started
root@ip-172-31-25-147:~/wordpress-compose#

```

### ➤ Verify Running Containers

```

root@ip-172-31-25-147:~/wordpress-compose# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
bc915c822370   wordpress:latest "docker-entrypoint.s..." 51 seconds ago Up 51 seconds 0.0.0
42cf55b0ceb7   mysql:5.7      "docker-entrypoint.s..." 54 seconds ago Up 51 seconds 3306/
root@ip-172-31-25-147:~/wordpress-compose#

```

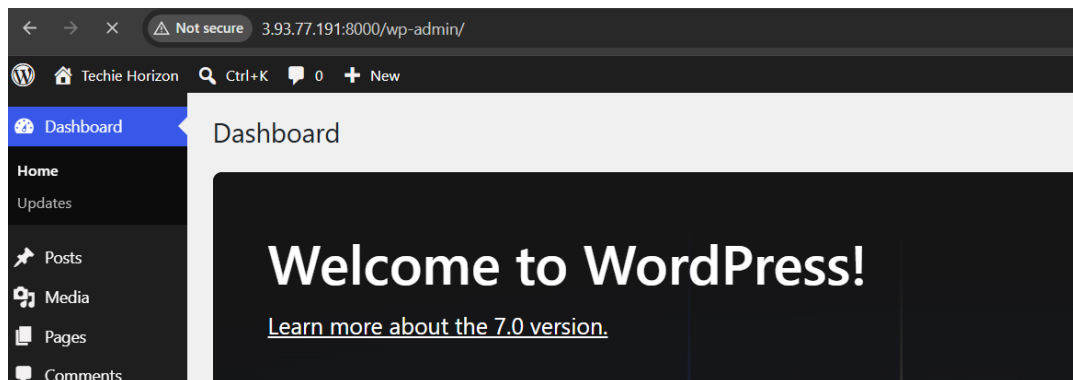
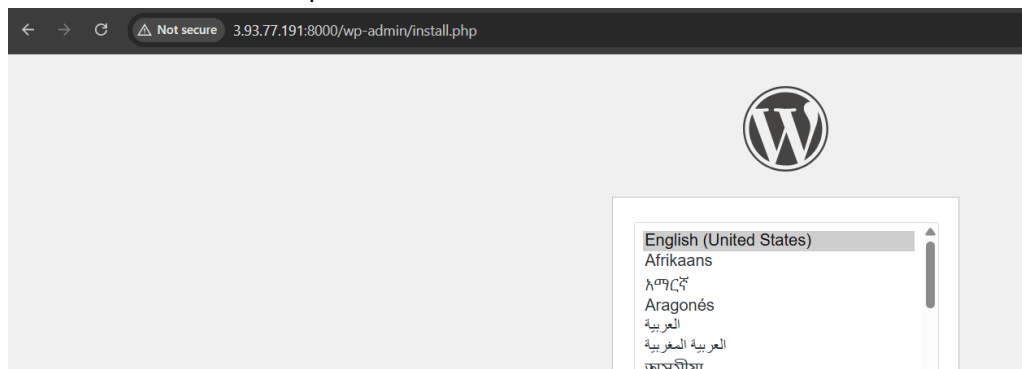
### ➤ Verify Docker Compose Services

```

root@ip-172-31-25-147:~/wordpress-compose# docker compose ps
WARN[0000] /root/wordpress-compose/docker-compose.yml: the attribute `version`
NAME                                IMAGE          COMMAND                  SE
wordpress-compose-db-1             mysql:5.7      "docker-entrypoint.s..." db
wordpress-compose-wordpress-1      wordpress:latest "docker-entrypoint.s..." wo
root@ip-172-31-25-147:~/wordpress-compose#

```

### ➤ Access WordPress with port no 8000



### ➤ Verify Persistent Volumes

```
wordpress-compose-wordpress-1 wordpress:latest "docker-entrypoint.s..."
root@ip-172-31-25-147:~/wordpress-compose# docker volume ls
DRIVER      VOLUME NAME
local       35a0c425f96af075b50f18d62cff4084a27907dde57dc2cf2268b585c06013fe
local       6007cf431a0da2d328f1b4c2d36d05f9a9d4e884e9c04561ad4d686f249b884a
local       deployment
local       jenkins_home
local       tomcat-volume
local       wordpress-compose_db_data
root@ip-172-31-25-147:~/wordpress-compose#
```

### ➤ Stop Containers

```
root@ip-172-31-25-147:~/wordpress-compose# docker compose down
WARN[0000] /root/wordpress-compose/docker-compose.yml: the attribute `vers
[+] Running 3/3
✓ Container wordpress-compose-wordpress-1 Removed
✓ Container wordpress-compose-db-1 Removed
✓ Network wordpress-compose_default Removed
root@ip-172-31-25-147:~/wordpress-compose#
```

### ➤ Restart Containers

```
root@ip-172-31-25-147:~/wordpress-compose# docker compose up -d
WARN[0000] /root/wordpress-compose/docker-compose.yml: the attribute `version`
[+] Running 3/3
✓ Network wordpress-compose_default Created
✓ Container wordpress-compose-db-1 Started
✓ Container wordpress-compose-wordpress-1 Started
root@ip-172-31-25-147:~/wordpress-compose#
```

### ➤ View Logs

```
root@ip-172-31-25-147:~/wordpress-compose# docker compose logs
WARN[0000] /root/wordpress-compose/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it t
wordpress-1 | WordPress not found in /var/www/html - copying now...
wordpress-1 | Complete! WordPress has been successfully copied to /var/www/html
wordpress-1 | No 'wp-config.php' found in /var/www/html, but 'WORDPRESS_...' variables supplied; copying 'wp-config-docker.php' (
WORDPRESS_DB_PASSWORD WORDPRESS_DB_USER)
wordpress-1 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the
ress this message
wordpress-1 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the
ress this message
wordpress-1 | [Wed Jun 03 16:17:35.240823 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.67 (Debian) PHP/8.3.31 con
wordpress-1 | [Wed Jun 03 16:17:35.240922 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'apache2 -D FOREGROUND'
db-1 | 2026-06-03 16:17:34+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
db-1 | 2026-06-03 16:17:34+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
db-1 | 2026-06-03 16:17:34+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
db-1 | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
db-1 | 2026-06-03T16:17:35.183017Z 0 [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_
ee documentation for more details).
db-1 | 2026-06-03T16:17:35.184413Z 0 [Note] mysqld (mysqld 5.7.44) starting as process 1 ...
db-1 | 2026-06-03T16:17:35.187206Z 0 [Note] InnoDB: PUNCH HOLE support available
db-1 | 2026-06-03T16:17:35.187270Z 0 [Note] InnoDB: Mutexes and rw_locks use GCC atomic builtins
db-1 | 2026-06-03T16:17:35.187282Z 0 [Note] InnoDB: Uses event mutexes
db-1 | 2026-06-03T16:17:35.187290Z 0 [Note] InnoDB: GCC builtin __atomic_thread_fence() is used for memory barrier
db-1 | 2026-06-03T16:17:35.187298Z 0 [Note] InnoDB: Compressed tables use zlib 1.2.13
db-1 | 2026-06-03T16:17:35.187308Z 0 [Note] InnoDB: Using Linux native AIO
```

## ERROR FILE

### ➤ I destroy the previous terraform and done changes main and userdata files

```
apply complete! Resources: 2 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform output

Warning: No outputs found

The state file either has no outputs defined, or all the defined outputs are empty. Please define
`terraform refresh` for it to become available. If you are using interpolation, please verify the
console` command to assist.

ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform show | grep public_ip
  associate_public_ip_address = true
  public_ip                   = "52.55.215.67"
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform show | grep ami
  ami = "ami-0c02fb55956c7d316"
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform destroy
aws_security_group.jenkins_sg: Refreshing state... [id=sg-0ef17f60476faa122]
aws_instance.jenkins_server: Refreshing state... [id=i-0617f00e67312c299]

Terraform used the selected providers to generate the following execution plan. Resource actions are
- destroy

Terraform will perform the following actions:

# aws_instance.jenkins_server will be destroyed
- resource "aws_instance" "jenkins_server" {
  - ami = "ami-0c02fb55956c7d316" -> null
```

```
Destroy complete! Resources: 2 destroyed.
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ vi main.tf
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ vi userdata.sh
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ cat main.tf | grep ami
data "aws_ami" "ubuntu" {
  ami = data.aws_ami.ubuntu.id
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ cat main.tf | grep ami
data "aws_ami" "ubuntu" {
  ami = data.aws_ami.ubuntu.id
ubuntu@ip-172-31-25-147:~/terraform-jenkins$ terraform init
terraform plan
terraform apply

Error: Terraform encountered problems during initialisation, including problem
with the configuration, described below.

The Terraform configuration must be valid before initialization so that
Terraform can determine which modules and providers need to be installed.
```

➤ In New EC2 Jenkins install manually but it fail

```
sudo apt install jenkins -y
--2026-06-04 09:53:21-- https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3175 (3.1K) [application/octet-stream]
Saving to: '/tmp/jenkins-keyring.asc'

/tmp/jenkins-keyring.asc          100%[=====]

2026-06-04 09:53:21 (24.7 MB/s) - '/tmp/jenkins-keyring.asc' saved [3175/3175]

Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:5 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Hit:7 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg
Reading package lists... Done
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures could
F4B714ABFC68
E: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
```

```
this may mean that the package is missing, has been moved, or
is only available from another source

E: Package 'jenkins' has no installation candidate
ubuntu@ip-172-31-22-87:~$ systemctl list-unit-files | grep jenkins
ubuntu@ip-172-31-22-87:~$ sudo systemctl enable jenkins
Failed to enable unit: Unit file jenkins.service does not exist.
ubuntu@ip-172-31-22-87:~$
ubuntu@ip-172-31-22-87:~$ dpkg -l | grep jenkins
ubuntu@ip-172-31-22-87:~$ cat /etc/apt/sources.list.d/jenkins.list
deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
ubuntu@ip-172-31-22-87:~$ sudo tail -100 /var/log/cloud-init-output.log
Adding debian:Trustwave_Global_ECC_P256_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P384_Certification_Authority.pem
```

```
sudo apt install jenkins -y
--2026-06-04 09:57:11-- https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:78::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3175 (3.1K) [application/octet-stream]
Saving to: '/tmp/jenkins-keyring.asc'

/tmp/jenkins-keyring.asc          100%[=====]

2026-06-04 09:57:11 (25.8 MB/s) - '/tmp/jenkins-keyring.asc' saved [3175/3175]

deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable bina
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:5 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Hit:7 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2056 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1696
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [32
Ign:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg
Reading package lists... Done
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures coul
F4B714ABFC68
```

```

E: Package 'jenkins' has no installation candidate
ubuntu@ip-172-31-22-87:~$ sudo systemctl enable jenkins
sudo systemctl start jenkins
sudo systemctl status jenkins
Failed to enable unit: Unit file jenkins.service does not exist.
Failed to start jenkins.service: Unit jenkins.service not found.
Unit jenkins.service could not be found.
ubuntu@ip-172-31-22-87:~$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | \
sudo gpg --dearmor -o /usr/share/keyrings/jenkins-keyring.gpg
ubuntu@ip-172-31-22-87:~$ echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.gpg] https://pkg.jenkins.io/debian-stable binary/" > /etc/apt/sources.list.d/jenkins.list
ubuntu@ip-172-31-22-87:~$ sudo rm -rf /var/lib/apt/lists/*
ubuntu@ip-172-31-22-87:~$ sudo apt update

```

```

2026-06-04 10:03:25 (11.1 KB/s) - 'index.html' saved [23458]

```

```

ubuntu@ip-172-31-22-87:~$ sudo dpkg -i jenkins_2.555_all.deb
dpkg: error: cannot access archive 'jenkins_2.555_all.deb': No such file or directory
ubuntu@ip-172-31-22-87:~$ sudo apt --fix-broken install -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 7 not upgraded.
ubuntu@ip-172-31-22-87:~$ sudo systemctl status jenkins
Unit jenkins.service could not be found.
ubuntu@ip-172-31-22-87:~$ grep ".deb" index.html | tail -20
<a href="jenkins_1.509.4_all.deb">jenkins_1.509.4_all.deb</a> 2013-10-09 17:59 50M
<a href="jenkins_1.509.3_all.deb">jenkins_1.509.3_all.deb</a> 2013-09-10 03:49 46M
<a href="jenkins_1.509.2_all.deb">jenkins_1.509.2_all.deb</a> 2013-06-28 06:10 46M
<a href="jenkins_1.509.1_all.deb">jenkins_1.509.1_all.deb</a> 2013-05-02 00:44 46M

```